Cell biology study notes. A cell is the basic structural and functional unit of life. The cell membrane consists of a phospholipid bilayer and regulates transport into and out of a cell. The nucleus contains DNA, which stores genetic information. Ribosomes synthesize proteins from messenger RNA. Mitochondria produce ATP during cellular respiration. ATP is an energy carrier used by cells. Chloroplasts perform photosynthesis in plant cells, converting light energy into chemical energy. The rough endoplasmic reticulum has ribosomes and helps produce proteins. The smooth endoplasmic reticulum synthesizes lipids. The Golgi apparatus modifies and packages proteins. Lysosomes contain enzymes that digest waste. Plant cell walls are mainly made of cellulose. Osmosis is movement of water across a selectively permeable membrane. Diffusion is movement of particles from higher concentration to lower concentration. Active transport requires energy to move substances against a concentration gradient.